Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code :-21317

F-417

B.C.A. (Fourth Semester)

Examination, 2024

(New Course)

Paper No. : BCA-N-402

Introduction to DBMS and SQL

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Explain the steps of normalization in Brief.

   (b) What is data model? Why is it important. Discuss advantages and disadvantages of data model.

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2. (a) Explain the concept of join Dependency with example in detail.

   (b) Explain Primary key, Foreign key, candidate Key, Super key & Composite key.

3. (a) Explain the different types of defenec mechanism.

   (b) What do you mean by data independence. Explain the difference between logical and physical data independence.

4. (a) Differentiate between DDL and DML in detail.

   (b) What are the anomalies of Third Normal Form (3NF)?

5. (a) What are the anomalies which leads to the bad design of database.

   (b) What is database Security? Why is database security important for an organization.

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6. (a) Discuss the comparison and contrast between a relation in BCNF and 3NF.

(b) Relational Model posses data independence. Explain.

7. (a) What are the features of data in the database? What operations can be performed on a database?

(b) Discuss history of Oracle. What are the various tools provided by the oracle?

8. (a) What is functional dependency? Explain with example.

(b) How do we convert an unnormalized relation to that in 1NF? What are the advantages and disadvantages of Normalization?

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